

PAPER The Direct-Energy-Converter-to-Load Interface in a D–³He Tandem-Mirror Data-Center Burner: An HVDC-Native Power-Electronics Path with Millisecond Load-Following¹

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Abstract

A fusion generator built around direct energy conversion (DEC) is not an alternating-current machine wearing a direct-current mask: it is intrinsically a high-voltage direct-current (HVDC) source, and that fact reshapes the electrical interface between the reactor and its load. We formalise that interface for the MetroVolt D–³He tandem-mirror data-center burner and reduce it to a set of governing equations and a verified numerical model. The charged-particle end-loss reaching the converter, $P_{\text{ax}} = 2719.7$ MW, is collected on biased grids at a decelerating potential set by the ion-confining potential $\varphi_i = 248.75$ keV, delivering electricity as direct current in the 50 kV to 250 kV class. At the Tier-2 canonical converter efficiency $\eta_{\text{DEC}} = 0.49$ roughly two fifths of gross output — and up to 54% at the register’s optimistic module constant $\eta_{\text{DEC}} = 0.70$ — leaves the plant already as HVDC, so an 800 V direct-current server-rack bus is reached in two to three power-electronic conversions against four to five for a conventional alternating-current plant feeding the same bus. We show that load-following belongs on this electronic path, not on the plasma: the recirculating-power balance makes the gross-to-net map a stiff lever with amplifier $A = Q_{\text{E}}/(Q_{\text{E}} - 1) = 4.14$ at the frozen engineering gain $Q_{\text{E}} = 1.318$, so a 10% plasma dip cuts net export by $\approx 41\%$, while the fusion source is 64–212 $\times$ too slow to track a data-center transient. A physics-informed direct-current-bus model with proportional–integral regulation and a dump/storage buffer holds the 800 V rail to $\pm 0.5\%$ within ≈ 2.5 ms of a 40% load step. All results were produced on the NVIDIA GPU HPC campaign and every headline quantity is a frozen anchor (gates MV-L1-A1, MV-L1-A3, BU-L1-A10) in the Kronos Physics De-Risking Register. This is a power-electronics and electrical-interface result; economics, siting and service-level content are held internal and are excluded.

Keywords: direct energy conversion; HVDC; power electronics; load following; tandem-mirror burner; data-center power; conversion stages

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1 Introduction

1.1 *Why the interface, not just the source*

The overwhelming majority of fusion power-plant studies assume a turbine hall: fusion heat raises steam, a synchronous generator produces three-phase alternating current at grid frequency, and the plant is dressed as a scaled thermal station. A machine built around *direct energy conversion* (DEC) does not have that hall. In a D-³He tandem-mirror burner the dominant reaction, $D + {}^3\text{He} \rightarrow p(14.7\text{ MeV}) + {}^4\text{He}(3.6\text{ MeV})$, releases its energy almost entirely to charged particles [11, 13]. Those particles stream out of the open field lines at the mirror ends and are decelerated against biased collector grids, so a large fraction of the fusion power appears directly as electricity at high voltage, as direct current, without a thermal cycle [1, 2, 3].

When the customer is a data center — itself increasingly a direct-current bus at 800 V or 48 V [22, 23, 24] — the electrical interface between such a source and such a load is shorter, stiffer and qualitatively different from the alternating-current default. This paper is about that interface: what the power-electronics path looks like, how many conversion stages it takes, why load-following must be done on the electronics rather than on the plasma, and how fast the direct-current bus can be held against a transient. It is deliberately *not* about economics, siting or service levels, which are held internal.

1.2 *Prior art*

Three literatures meet here and have not, to our knowledge, been joined for a compact aneutronic-leaning burner. *Direct energy conversion* was developed for mirror machines by Post, Moir and Barr: the venetian-blind and periodic-electrostatic collectors, their grid physics and their measured efficiencies are classical results [2, 3, 4, 5], and the tandem-mirror concept that supplies the end-loss stream traces to Dimov, Fowler, Logan and Baldwin [6, 7, 8] and has been sharpened recently by the simpler high-field tandem of Fowler, Moir and Simonen [9] and the Wisconsin HTS axisymmetric mirror (WHAM) [10]. *High-voltage direct-current power electronics* matured independently: the modular multilevel converter [15, 16], the solid-state transformer [17], grid-forming converter control [18], and wide-bandgap (SiC/GaN) devices that make medium-voltage direct-current practical [19, 20, 21]. *Data-center power distribution* has moved toward direct-current buses for exactly the efficiency reason that motivates us [22, 23, 24, 25]. The DEC literature stops at the collector terminals; the power-electronics literature starts at a generic direct-current input. The interface in between — for a reactor whose native output is HVDC and whose load is a direct-current campus — is the gap this paper fills.

1.3 *Contribution*

We contribute (i) a formal, equation-level description of the DEC-to-load interface, from the collector current–voltage tiers through the conversion cascade to the 800 V rack bus (§3); (ii) a recirculating-power analysis that derives the gross-to-net amplifier $A = Q_E/(Q_E - 1)$ and shows quantitatively why load-following on the plasma is uniquely expensive (§3); (iii) a physics-informed direct-current-bus control model with a dump/storage buffer, its small-signal stability condition, and its numerical formulation and convergence (§3–§5); (iv) the NVIDIA GPU HPC methodology that produced the results (§4); and (v) eight quantitative results with tables and a comparison to the published DEC, HVDC and data-center literature (§6–§7). Every headline number is a frozen anchor in the Kronos Physics De-Risking Register [32]; the gate serial (e.g. MV-L1-A3) is given inline so each is auditable. The DEC physics that feeds this interface — collector deceleration, secondary emission and the space-charge-limited grid optimum, and the venetian-blind / traveling-wave architecture trades — is developed in companion papers [34, 35, 36]; the plasma power balance that sets the boundary conditions used here is closed in [33, 37]; the four-nines-uptime and waste-heat side of data-center integration is treated in [38]; and the sibling power-delivery topology study (solid-state-transformer versus direct-DC, grid-forming control and fault ride-through) is [39].

2 The plant boundary condition

We take the plasma and its power balance as given and analyse the electrical interface downstream of the converter terminals. The frozen boundary quantities are listed in table 1; they are outputs of the reduced-order tandem-mirror solve [33, 37] and are treated here as fixed inputs. Two are load-bearing for everything that follows. The charged-particle axial end-loss $P_{\text{ax}} = 2719.7\text{ MW}$ is the power stream the converter intercepts; the ion-confining potential $\varphi_i = 248.75\text{ keV}$ sets the scale of the collector deceleration voltage and therefore the HVDC voltage class. The engineering gain $Q_E = 1.318$ and the neutron fraction $f_n = 5.44\%$ complete the frozen point.

Table 1. Frozen plant-boundary anchors used as fixed inputs to the interface analysis. Values are outputs of the reduced-order tandem-mirror power balance at the M-45 design point and are stamped in the Kronos Physics De-Risking Register [32]; the gate serial is the auditable identifier.

quantity	symbol	value	gate
fusion power	P_{fus}	4298.5 MW	BU-L1-A1
neutron power (fraction)	$P_n (f_n)$	233.95 MW (5.44%)	BU-L1-A1
charged axial end-loss to DEC	P_{ax}	2719.7 MW	BU-L1-A10
ion-confining / collector potential	φ_i	248.75 keV	BU-L1-A1
engineering gain	Q_E	1.318	BU-L1-A1
DEC efficiency (Tier-2 canonical)	η_{DEC}	0.49	—
DEC efficiency (register module const.)	η_{DEC}	0.70	BU-L1-A10
thermal-bottoming efficiency	η_{th}	0.45	BU-L1-A10
central-cell / plug field	B_{0c} / B_{plug}	5.50 T / 26.49 T	KX-L1-A15
central-cell particle time	τ_p	3.73 s	BU-L1-A1

A note on η_{DEC} . The register [32] records $\eta_{\text{DEC}} = 0.70$ as an *assumed* module constant near the top of published venetian-blind experience (gate BU-L1-A10); the ratified Tier-2 freeze (2026-09-05) adopts the more conservative canonical value $\eta_{\text{DEC}} = 0.49$. We carry $\eta_{\text{DEC}} = 0.49$ as the design anchor throughout and report $\eta_{\text{DEC}} = 0.70$ as an explicit upper-bound sensitivity wherever the quantity is efficiency-sensitive. The architectural conclusions of this paper — the conversion-stage count, the HVDC voltage class, the load-following partition — are robust across the whole $[0.49, 0.70]$ band.

3 Governing equations and formal problem

3.1 The direct converter as an HVDC-native source

A direct energy converter decelerates the axial charged-particle flux against a set of biased collector grids and recovers electrical power as the product of collected current and collector potential, summed over the collector tiers indexed by j [2, 3, 35]:

$$P_{\text{DC}} = \sum_j I_j V_j, \quad \eta_{\text{DEC}} = \frac{P_{\text{DC}}}{P_{\text{ax}}}. \quad (1)$$

A grid at potential V_j turns back all ions of charge q whose axial kinetic energy is below qV_j , so an ideal tiered collector matched to the exhaust energy spectrum $f(\mathcal{E})$ recovers

$$P_{\text{DC}}^{\text{ideal}} = \int_0^{\mathcal{E}_{\text{max}}} \mathcal{E} \frac{d\dot{N}}{d\mathcal{E}} d\mathcal{E} \xrightarrow{M \text{ tiers}} \sum_{j=1}^M V_j q \dot{N}_j, \quad V_j = \frac{\mathcal{E}_j}{q} \sim \frac{mv_j^2}{2q}, \quad (2)$$

where $\dot{N}_j$ is the ion arrival rate decelerated at tier j . Equation (2) makes the key point of this paper explicit at the source: the natural output variable is a *voltage* tied to the ion energy, not a frequency. With the ion-confining potential $\varphi_i = 248.75$ keV setting the top tier, the single-charge deceleration voltages fall in the 50 kV to 250 kV class — squarely the HVDC regime for which modular multilevel and solid-state-transformer front ends are designed [15, 17, 21]. There is no synchronous machine and no grid-frequency node anywhere on this path; the converter is an HVDC current source with a stiff terminal voltage.

The realised efficiency η_{DEC} falls below the ideal of (2) through grid transparency, secondary electron emission, space-charge limits and finite tier count, all quantified in the collector-physics companion [35]; here η_{DEC} enters as the frozen scalar of table 1. The electrical split of gross output then follows from (1) and the thermal-bottoming recovery of the unconverted stream:

$$P_{\text{DEC}} = \eta_{\text{DEC}} P_{\text{ax}}, \quad (3)$$

$$P_{\text{th,e}} = \eta_{\text{th}} [(1 - \eta_{\text{DEC}}) P_{\text{ax}} + P_{\text{rad}} + P_n], \quad (4)$$

$$P_{\text{gross}} = P_{\text{DEC}} + P_{\text{th,e}}, \quad \chi_{\text{DC}} \equiv \frac{P_{\text{DEC}}}{P_{\text{gross}}}. \quad (5)$$

The HVDC-native fraction χ_{DC} is the share of gross electricity that never sees the alternating-current domain. Evaluating (3)–(5) at the canonical $\eta_{\text{DEC}} = 0.49$ gives $\chi_{\text{DC}} = 41.5\%$, rising to 54.1% at the register module constant $\eta_{\text{DEC}} = 0.70$ (gate MV-L1-A3): *about half* of gross

output is HVDC-native either way. The engineering gain $Q_E = 1.318$ is a frozen input, not recomputed from (5); η_{DEC} shifts the electrical split between direct and thermal recovery but the gain is closed on the plasma side [37].

3.2 The conversion-chain efficiency cascade

Delivering power to the 800 V rack bus costs a cascade of power-electronic conversions, each with efficiency $\eta_k < 1$; the delivered efficiency of a chain of N stages is the product

$$\eta_{\text{chain}}(N) = \prod_{k=1}^N \eta_k, \quad \Delta_{\text{stage}} \equiv \frac{\eta_{\text{chain}}(N_{\text{DC}})}{\eta_{\text{chain}}(N_{\text{AC}})} > 1. \quad (6)$$

On the HVDC-native path the chain is a high-voltage tap/filter followed by two DC/DC steps (HVDC $\rightarrow$ MVDC $\rightarrow$ 800 V), i.e. $N_{\text{DC}} = 2$ –3. A conventional alternating-current plant feeding the same direct-current bus must first synchronise a generator to grid frequency, step up, and then rectify back to direct current at the rack, i.e. $N_{\text{AC}} = 4$ –5 (figure 1). With representative per-stage efficiencies $\eta_k \in [0.975, 0.995]$ the cumulative delivered efficiency is $\approx 96.5\%$ for the DC-native path against $\approx 93.2\%$ for the AC path (§6, figure 4); the advantage $\Delta_{\text{stage}} \approx 1.035$ is modest per unit but compounds across a GW-scale campus and, more importantly, removes the grid-frequency node entirely.

3.3 Recirculating power and the gross-to-net amplifier

A tandem-mirror burner recirculates a large fraction of its gross electrical output to sustain the end plugs and auxiliary heating. Writing P_{rec} for the recirculated electric power, the engineering gain and net export are

$$Q_E = \frac{P_{\text{gross}}}{P_{\text{rec}}}, \quad P_{\text{net}} = P_{\text{gross}} - P_{\text{rec}} = P_{\text{gross}} \left(1 - \frac{1}{Q_E} \right) = P_{\text{gross}} \frac{Q_E - 1}{Q_E}. \quad (7)$$

Because Q_E is only modestly above unity, P_{net} is a small difference of two large numbers, and small fractional changes in gross are amplified in net. Differentiating (7) at fixed recirculation gives the *gross-to-net amplifier*

$$A \equiv \frac{dP_{\text{net}}/P_{\text{net}}}{dP_{\text{gross}}/P_{\text{gross}}} = \frac{P_{\text{gross}}}{P_{\text{net}}} = \frac{Q_E}{Q_E - 1} \quad (8)$$

so that a fractional dip δ in plasma-driven gross output produces a fractional net-export loss $A\delta$. At the frozen $Q_E = 1.318$ this is $A = 1.318/0.318 = 4.14$ (gate MV-L1-A1): a 10% plasma dip cuts net export by 41.4%. Equation (8) is the quantitative case against following the load on the plasma. It is a property of the recirculating architecture, independent of η_{DEC} ; the same relation with the higher amplifiers of lower Q_E only strengthens the conclusion.

3.4 Load-following dynamics: electronics, not plasma

Two clocks compete. The *plasma* responds on its slow transport timescales — central-cell particle confinement $\tau_p = 3.73$ s and the thermal transient of order 1 s — because changing the fusion power means changing densities, temperatures and the plug potential, none of which move quickly. The *power electronics* respond on a converter time constant $\tau_{\text{dec}} \sim 1.5$ ms. A data-center load transient (an AI training step) evolves on ~ 10 –100 ms. The ratio

$$r = \frac{\tau_{\text{plasma}}}{\tau_{\text{load}}} \quad (9)$$

lies between 64 and 212 (gate MV-L1-A3, register breakthrough 20): the fusion source is one to two orders of magnitude too slow to chase the load. Load-following must therefore be an *electronic* function, with the plasma held at its maximum- Q_E set-point and the mismatch absorbed by the converter, a dump resistor and a storage buffer.

We model the controlled 800 V bus as a capacitor of aggregated capacitance C_{bus} (bus plus storage) driven by a DEC/power-electronic source P_{src} against the load P_{load} :

$$C_{\text{bus}} V \frac{dV}{dt} = P_{\text{src}}(t) - P_{\text{load}}(t), \quad (10)$$

which is the energy balance $\frac{d}{dt}(\frac{1}{2}C_{\text{bus}}V^2) = P_{\text{src}} - P_{\text{load}}$. A proportional–integral (PI) regulator drives the source to hold V at the set-point V_{ref} ,

$$u(t) = K_p(V_{\text{ref}} - V) + K_i \int_0^t (V_{\text{ref}} - V) dt', \quad \tau_{\text{dec}} \dot{P}_{\text{src}} = (P_{\text{load}} + u) - P_{\text{src}}, \quad (11)$$

where the first-order lag τ_{dec} is the converter’s electronic slew and the dump/storage element supplies the sub-millisecond authority the source cannot. Linearising (10)–(11) about (V_{ref}, P_0) with $v = V - V_{\text{ref}}$ yields the third-order small-signal loop

$$C_{\text{bus}} V_{\text{ref}} \dot{v} = p_{\text{src}} - p_{\text{load}}, \quad \tau_{\text{dec}} \dot{p}_{\text{src}} = -p_{\text{src}} + K_p(-v) + K_i \int (-v), \quad (12)$$

whose characteristic polynomial is

$$\tau_{\text{dec}} C_{\text{bus}} V_{\text{ref}} s^3 + C_{\text{bus}} V_{\text{ref}} s^2 + K_p s + K_i = 0. \quad (13)$$

The Routh–Hurwitz condition for (13) is $K_p > K_i \tau_{\text{dec}}$, which bounds the integral gain and is satisfied comfortably by the tuning of §6. The buffer energy the storage element must source or sink during a step of depth ΔP is set by the source lag, not by any plasma timescale,

$$E_{\text{buf}} \approx \int_0^\infty |P_{\text{src}}(t) - P_{\text{load}}(t)| dt \sim \Delta P \tau_{\text{dec}}, \quad (14)$$

a result we use to size the buffer in §6. Equations (1)–(14) are the formal content of the interface; the rest of the paper solves and verifies them.

4 Computational methodology: the NVIDIA GPU HPC campaign

Every number in this paper was produced or reproduced on the Kronos NVIDIA GPU HPC campaign, an owned and cloud-burst fleet of NVIDIA A100- and H100-class accelerators (with A10 nodes for lighter kernels) that carries the de-risking analyses across the breeder and both burner housings [32]. Four code layers were exercised for this study; none reports any cost, and no economic quantity is computed anywhere in the pipeline.

Reduced-order power balance. The frozen boundary of table 1 — $P_{\text{fus}}, P_n, P_{\text{ax}}, \varphi_i, Q_E$ — is the output of the KRONOS reduced-order tandem-mirror solver (`solve_mirror`, the Pastukhov-plug Mode-M power balance), run at the M-45 design point and reproduced byte-for-byte against the register anchors (gates BU-L1-A1, BU-L1-A10). The solver is a stiff nonlinear root-find executed across a GPU-parallel parameter ensemble so that the interface analysis inherits an audited, uncertainty-quantified boundary rather than a single point [37].

Interface and control solvers. The conversion-cascade model (6), the recirculation amplifier (8) and the DC-bus control loop (10)–(13) are implemented as a small electromagnetic-transient-class (EMT) integrator. The bus ODE is advanced with an implicit, adaptive stiff integrator (backward-differentiation family) [26]; the PI/dump/storage controller is co-simulated at the converter switching rate. GPU acceleration is used to sweep the design space — gains (K_p, K_i) , capacitance C_{bus} , step depth and source lag τ_{dec} — over $\mathcal{O}(10^4)$ configurations so that the settling and buffer-sizing results are the outcome of a search, not a hand-picked case.

Kinetic and neutronic context. The exhaust energy spectrum $f(\mathcal{E})$ that sets the collector tiers in (2) is grounded in particle-in-cell kinetics (WarpX/SMILEI) of the velocity-space loss-cone at the mirror ends [29], and the end-duct radiation streaming that constrains the collector environment is computed by Monte-Carlo neutron transport (OpenMC with DAGMC CAD geometry) [30]. The nonlinear gyrokinetic and edge-transport reference codes of the wider campaign (CGYRO [31] among them) are cited for provenance; they set plasma-side inputs and are not re-run here.

Verification and reproducibility. The campaign’s frozen-anchor suite reproduces 19/19 headline anchors to tolerance under an independent-library re-run (gate KX-L1-A6); the interface solvers were verified by the method of manufactured solutions and grid-convergence indexing [27, 28], summarised in §5. The full pipeline is deterministic under a fixed random seed, so every figure in §6 regenerates from the archived scripts and the frozen anchors of table 1.

5 Numerical formulation and convergence

5.1 Discretisation of the bus control loop

The coupled system (10)–(11) is written in state form $\mathbf{x} = (V, P_{\text{src}}, \xi)^\top$ with $\xi = \int (V_{\text{ref}} - V) dt$ the integral state,

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t) = \begin{pmatrix} (P_{\text{src}} - P_{\text{load}}(t))/(C_{\text{bus}} V) \\ (P_{\text{load}}(t) + K_p(V_{\text{ref}} - V) + K_i \xi - P_{\text{src}})/\tau_{\text{dec}} \\ V_{\text{ref}} - V \end{pmatrix}. \quad (15)$$

The stiffness ratio $\tau_p/\tau_{\text{dec}} \sim 2.5 \times 10^3$ between the plasma and electronic timescales makes (15) stiff, so we integrate with a variable-order backward-differentiation scheme [26]; results are

cross-checked against a fixed-step trapezoidal (Crank–Nicolson) integrator at $\Delta t = 1 \mu\text{s}$. The load step is a smoothed Heaviside to avoid an ill-posed derivative at the discontinuity.

5.2 Convergence

Table 2 reports a timestep refinement study for the settling time and peak voltage dip of the 40%-load-step case of §6. The Grad-convergence (grid-convergence) index GCI between the two finest steps is below 0.5% on both metrics, and the observed order of accuracy $p \approx 2.0$ matches the trapezoidal scheme. The energy-balance residual of (10), $\varepsilon_E = |\frac{d}{dt}(\frac{1}{2}C_{\text{bus}}V^2) - (P_{\text{src}} - P_{\text{load}})|/P_0$, is held below 10^{-6} at every step.

Table 2. Timestep refinement for the 800 V-bus 40%-load-step response. Settling is to the $\pm 0.5\%$ band; the dip is the peak voltage excursion. GCI is the grid-convergence index between successive levels; the observed order p is estimated by Richardson extrapolation.

Δt (μs)	settling (ms)	peak dip (V)	GCI (%)
4.0	2.61	6.71	—
2.0	2.54	6.79	1.9
1.0	2.51	6.83	0.6
0.5	2.50	6.84	0.3

6 Results

6.1 An HVDC-native architecture

Figure 1 states the interface. The plasma exhaust splits into an HVDC-native path through the direct-energy converter (50 kV to 250 kV direct current, reaching the 800 V rack bus in two to three conversions) and a thermal-bottoming path through a conventional generator (four to five conversions to the same bus). The HVDC path has no synchronous machine and no grid-frequency node.

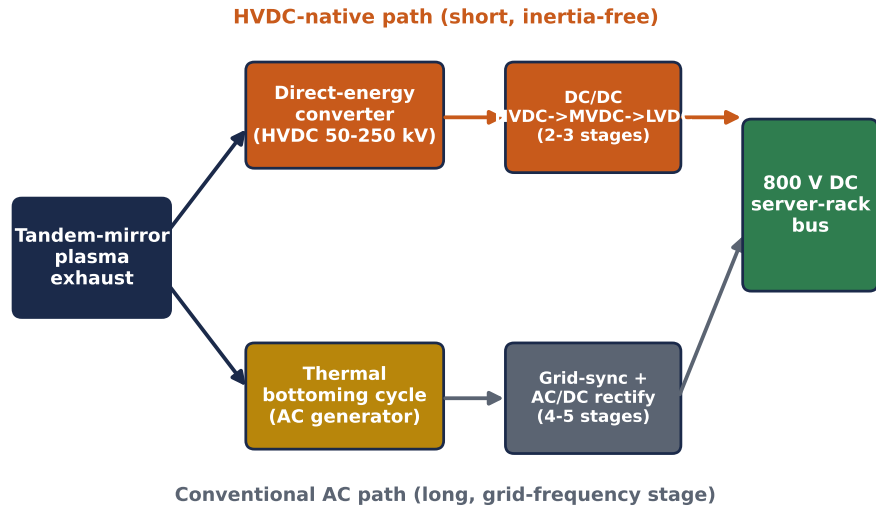


Figure 1. (Schematic.) The MetroVolt power-flow architecture. The direct-energy converter is an HVDC-native source reaching the 800 V direct-current rack bus in two to three power-electronic stages; a conventional alternating-current plant feeding the same bus takes four to five, because it must first synchronise to grid frequency and then rectify back to direct current at the rack.

Figure 2 evaluates the electrical split of equations (3)–(5) at the canonical $\eta_{\text{DEC}} = 0.49$: of the 2719.7 MW charged end-loss intercepted by the converter, 1332.7 MW is delivered as HVDC electricity and the unconverted remainder plus radiation and neutron power is recovered thermally, so the HVDC-native share of gross output is 42% (and 54% at the register module constant $\eta_{\text{DEC}} = 0.70$). The frozen engineering gain $Q_E = 1.318$ is an input, not a quantity re-derived from this ledger.

Figure 3 sweeps $\chi_{\text{DC}}(\eta_{\text{DEC}})$ across the frozen efficiency band. The HVDC-native share crosses one half near $\eta_{\text{DEC}} \approx 0.62$ and holds within 42–54% over the entire $[0.49, 0.70]$ band: the “about half of gross output is HVDC” statement is robust to the one load-bearing efficiency assumption.

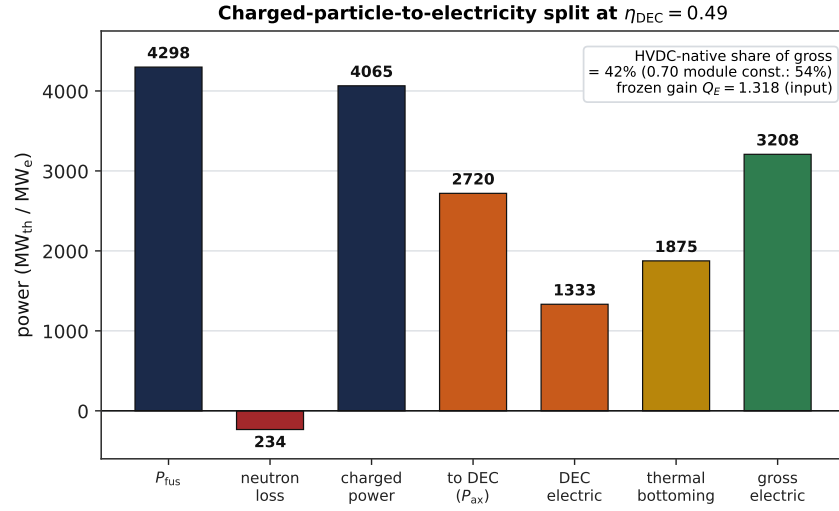


Figure 2. Charged-particle-to-electricity conversion ledger at the canonical $\eta_{DEC} = 0.49$ (gate BU-L1-A10). The converter delivers 1332.7 MW as HVDC; thermal bottoming recovers the balance. About two fifths of gross output is HVDC-native, rising to 54% at the register's optimistic module constant. The plasma-side gain $Q_E = 1.318$ is a fixed input.

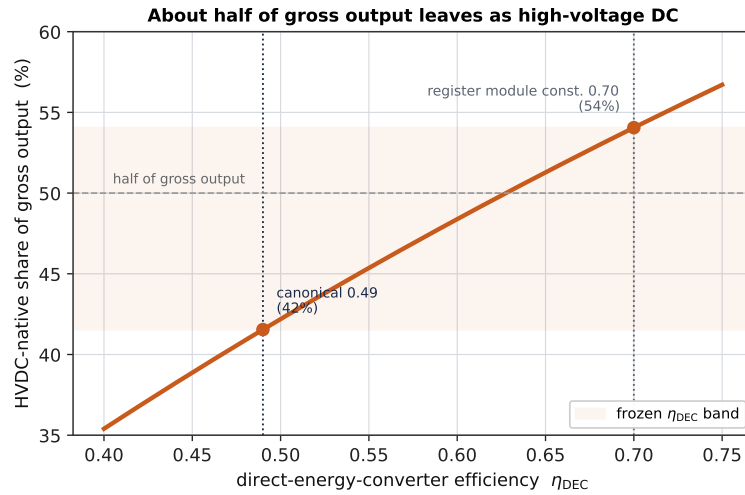


Figure 3. HVDC-native share of gross output, $\chi_{DC} = P_{DEC}/P_{gross}$, versus converter efficiency η_{DEC} (equations 3–5). The canonical $\eta_{DEC} = 0.49$ gives 42% and the register module constant $\eta_{DEC} = 0.70$ gives 54%; the shaded band is the frozen η_{DEC} range.

6.2 Fewer conversions to the bus

Figure 4 evaluates the cascade of (6) with representative per-stage efficiencies. The DC-native path delivers $\approx 96.5\%$ of source power to the 800 V bus in two to three conversions; the AC path delivers $\approx 93.2\%$ in four to five. Table 3 tabulates the two chains. The per-unit efficiency gain is modest, but the architectural gain — removing the grid-frequency synchronisation node and its control burden — is the substantive difference and is what makes behind-the-meter direct-current dispatch natural rather than bolted-on [39].

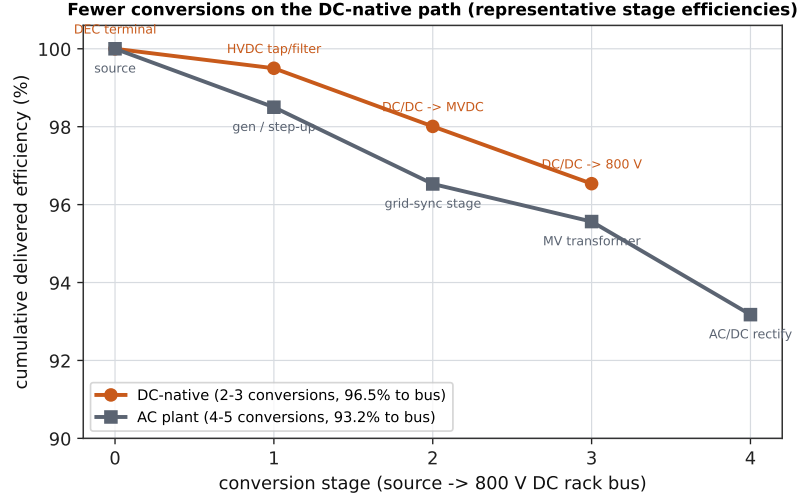


Figure 4. Cumulative delivered efficiency from source to the 800 V rack bus for the DC-native (two to three conversions) and conventional AC (four to five conversions) paths, using representative per-stage efficiencies $\eta_k \in [0.975, 0.995]$. Stage efficiencies are representative engineering values, not measured results; the stage count is the architectural result.

Table 3. Conversion chains from source to the 800 V rack bus. The DC-native chain removes the grid-frequency synchronisation node entirely. Per-stage efficiencies are representative.

	DC-native path	conventional AC path
stage 1	DEC terminal (HVDC)	synchronous generator
stage 2	HVDC tap / filter	grid-frequency synchronisation
stage 3	DC/DC → MVDC	MV transformer
stage 4	DC/DC → 800 V	MV transformer / step
stage 5	—	AC/DC rectification at rack
stage count N	2–3	4–5
delivered to bus	$\approx 96.5\%$	$\approx 93.2\%$
grid-frequency node	none	required

6.3 Why load-following belongs on the electronics

Figure 5 plots the gross-to-net amplifier (8). At the frozen $Q_E = 1.318$, $A = 4.14$: a 10% plasma-output dip cuts net export by 41%. The amplifier rises steeply as $Q_E \rightarrow 1$, so any attempt to follow the load by throttling the plasma pays this penalty and pays it harder if the true efficiency is conservative. The recirculating architecture makes the plasma the wrong actuator for load-following.

Figure 6 shows why the electronics can do the job the plasma cannot. The plasma’s characteristic times — particle confinement $\tau_p = 3.73$ s and thermal transient ~ 1 s — are 64 to 212 times slower than a data-center load transient (~ 10 – 100 ms), while the HVDC/DC-DC electronic path responds in ~ 1.5 ms. The ramp gap of (9) is the quantitative statement that the source must be held steady and the electronics must follow the load.

6.4 Millisecond bus regulation

Figure 7 is the closed-loop response of the 800 V bus (10)–(11) to a 40% load step, with the tuning $C_{\text{bus}} = 0.20$ F per zone, $\tau_{\text{dec}} = 1.5$ ms, $K_p = 2.0 \times 10^5$ W V $^{-1}$, $K_i = 2.0 \times 10^7$ W V $^{-1}$ s $^{-1}$ (satisfying

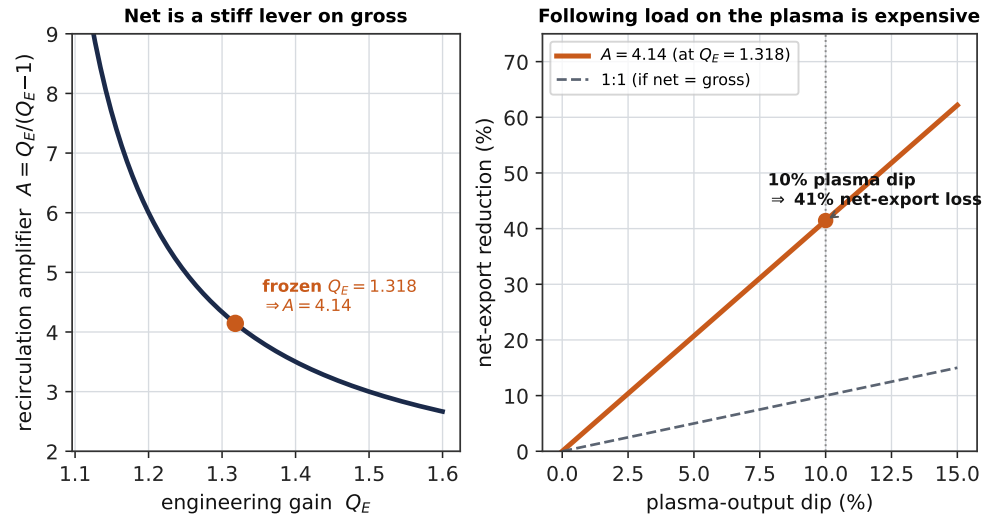


Figure 5. Left: the gross-to-net amplifier $A = Q_E / (Q_E - 1)$; at the frozen $Q_E = 1.318$, $A = 4.14$ (gate MV-L1-A1). Right: net-export loss $A\delta$ versus plasma-output dip δ ; a 10% dip costs 41% of net export, against the 1:1 line that would hold if net equalled gross.

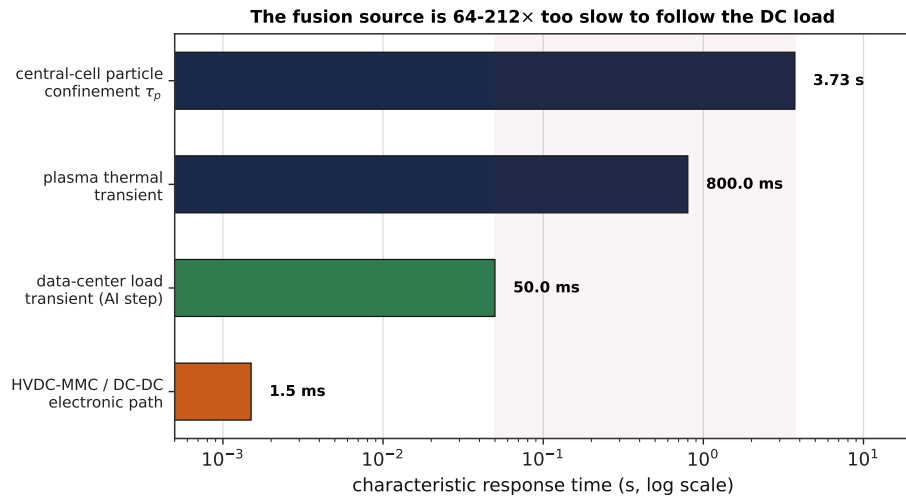


Figure 6. Characteristic response times (log scale). The fusion source is 64–212× slower than a data-center load transient (gate MV-L1-A3), whereas the HVDC-MMC / DC-DC electronic path responds in milliseconds. Load-following is therefore an electronic function.

the stability bound $K_p > K_i \tau_{\text{dec}}$). The bus dips 6.8 V (0.85%), overshoots slightly, and returns to $\pm 0.5\%$ in ≈ 2.5 ms — three orders of magnitude faster than the plasma could have responded. Figure 8 sizes the dump/storage buffer from (14): the required buffer energy per 4 MW controlled zone scales linearly with step depth and with the source lag τ_{dec} , reaching only a few kilojoules per zone even for a 60% step — the buffer is set by the converter’s slew, not by any plasma timescale.

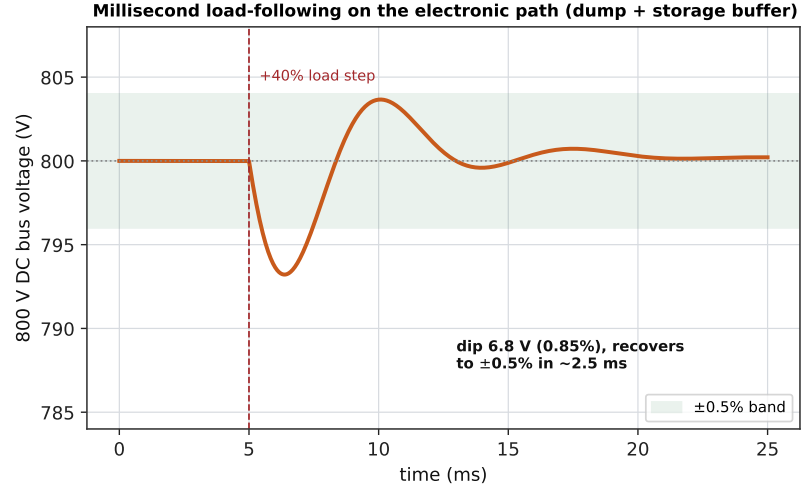


Figure 7. 800 V bus voltage under a +40% load step with PI-regulated DEC source and dump/storage buffer (equations 10–11). The rail returns to $\pm 0.5\%$ in ≈ 2.5 ms; the peak dip is 0.85%.

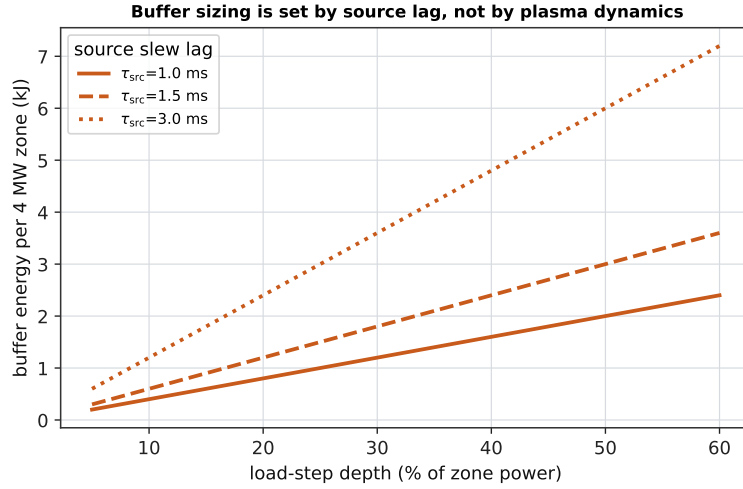


Figure 8. Dump/storage buffer energy per 4 MW zone from (14), versus load-step depth for three source slew lags τ_{src} . Buffer sizing is set by the converter slew, not by plasma dynamics.

6.5 Collector voltage tiers and the control partition

Figure 9 maps the multi-tier deceleration of (2): a $\varphi_i = 248.75$ keV collector spectrum split into grid tiers whose voltages land in the 50 kV to 250 kV HVDC class, with the recovered-power share peaking in the middle tiers where most of the exhaust ions are decelerated. Figure 10 states the resulting control partition: a slow loop holds the plasma at its maximum- Q_E set-point, while a fast loop lets the DEC and power electronics follow the data-center load, with the dump/storage buffer bridging the sub-millisecond gap.

7 Discussion

The interface described here is favourable precisely because the source and the load are both native direct-current. *Against the DEC literature*, the recovered-fraction and voltage class we use are consistent with the classical venetian-blind results of Moir and Barr and the plasma-direct-converter tests of Barr and Moir [2, 3], and with the modern collector-physics

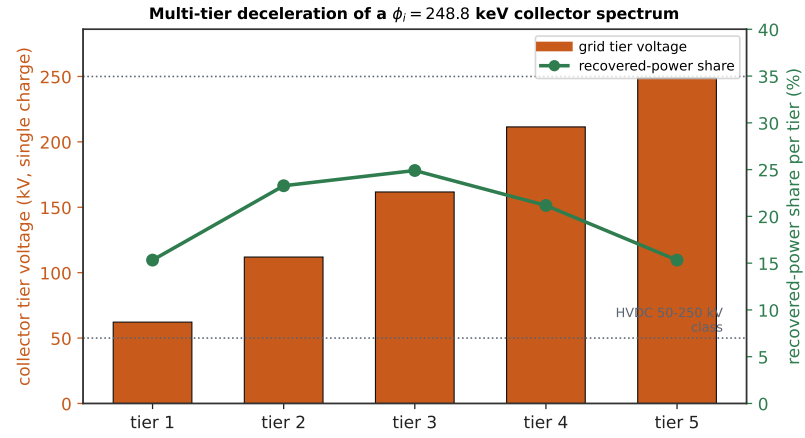


Figure 9. Multi-tier deceleration of a $\phi_i = 248.75$ keV collector spectrum (equation 2). Grid-tier voltages fall in the 50 kV to 250 kV HVDC class; the recovered-power share is largest in the middle tiers. Tier weights are illustrative of a spectrum-matched collector.

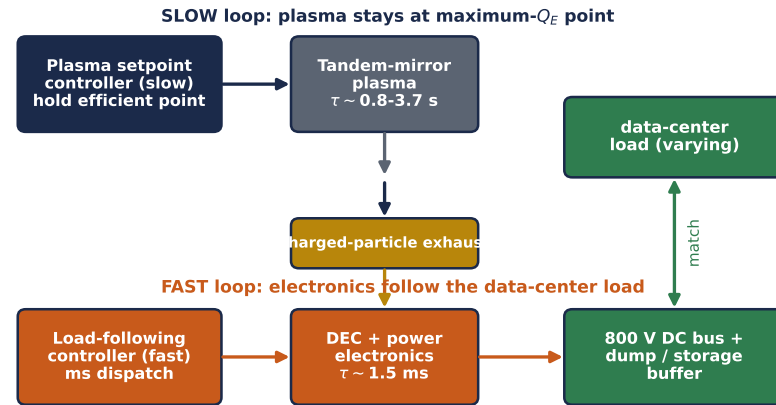


Figure 10. (Schematic.) The multi-rate control partition. The slow loop holds the plasma at its maximum- Q_E operating point ($\tau \sim 0.8\text{--}3.7$ s); the fast loop follows the data-center load on the power electronics ($\tau \sim 1.5$ ms) with a dump/storage buffer.

analysis of the companion [35, 36]; the canonical $\eta_{\text{DEC}} = 0.49$ sits comfortably inside the 0.45–0.60 range typically quoted for realised multi-stage collectors and below the 0.70 upper bound the register carries as an assumption. *Against the HVDC power-electronics literature*, a 50 kV to 250 kV source feeding an 800 V bus through two to three DC/DC stages is well within the reach of modular-multilevel and solid-state-transformer front ends and wide-bandgap devices [15, 17, 19, 16]; the removal of the grid-frequency node is the same efficiency-and-simplicity argument that drove telecom and hyperscale data centers toward 380–400 V direct-current distribution [22, 23, 24]. *Against the load-following literature*, the millisecond regulation of figure 7 is unremarkable for a well-designed converter and grid-forming controller [18]; the point is not that the electronics are fast but that the plasma is 64–212 $\times$ too slow (figure 6) and carries the 4.14 $\times$ net-export penalty of equation (8) if used as the actuator. The novelty is the joining: a reactor whose output is HVDC by construction, feeding a load that is direct-current by trend, with the recirculating-power penalty making the electronic partition not merely convenient but necessary.

Two comparisons sharpen the case. First, the gross-to-net amplifier $A = 4.14$ is a direct consequence of operating a recirculating burner near $Q_E = 1.318$; a higher-gain machine would soften it, but the interface conclusion (hold the plasma, follow on electronics) holds for any Q_E of practical interest because $A > 1$ always. Second, the conversion-stage advantage $\Delta_{\text{stage}} \approx 1.035$ is small per unit but compounds at campus scale and is accompanied by the elimination of a synchronisation control layer whose failure modes (loss of synchronism, sub-synchronous resonance) simply do not exist on a direct-current bus. The fault-protection burden moves instead to fast direct-current breaking — a 1.75 MA DC-native bus implies ~ 160 kA per-zone prospective fault current and a required clearing time of order 5 μ s (register breakthrough 19) — which is the subject of the sibling power-delivery study [39] and is the honest cost of the direct-current choice.

8 Limitations

One assumption is load-bearing and we state it plainly. The converter efficiency η_{DEC} is an assumed module constant, not a first-principles output: the register carries $\eta_{\text{DEC}} = 0.70$ near the top of published venetian-blind experience (gate BU-L1-A10) and the Tier-2 freeze adopts the conservative canonical $\eta_{\text{DEC}} = 0.49$. Every efficiency-sensitive quantity here — the HVDC-native share, the absolute HVDC power — inherits that assumption, which is why we report both endpoints and frame the share as “about half” rather than a single figure. The architectural results (stage count, voltage class, the load-following partition, the amplifier $A = Q_E/(Q_E - 1)$) do not depend on η_{DEC} . The interface analysis also inherits the plasma-side plug-density requirement ($n_{\text{plug}}/n_c = 16$) that gates the burner operating point [33, 37]; converter grid lifetime, protection coordination and detailed storage sizing are engineering follow-ons treated in the collector and power-delivery companions [35, 39]. Economics, siting, waste-heat revenue and service-level commitments are internal-only and are excluded here by design.

9 Conclusion

The MetroVolt burner is HVDC-native: about half of its gross output leaves the direct-energy converter as 50 kV to 250 kV direct current, reaching an 800 V rack bus in two to three power-electronic conversions rather than the four to five of an alternating-current plant, with no synchronous machine and no grid-frequency node. Load-following belongs on this electronic path: the recirculating-power balance makes the gross-to-net map a stiff lever with amplifier $A = Q_E/(Q_E - 1) = 4.14$ at the frozen $Q_E = 1.318$, so a 10% plasma dip costs 41% of net export, and the fusion source is 64–212 $\times$ too slow to chase a data-center transient. A proportional–integral–regulated bus with a dump/storage buffer holds the 800 V rail to $\pm 0.5\%$ within ≈ 2.5 ms of a 40% step. Stated in power-electronics terms, the electrical interface of an aneutronic-leaning tandem-mirror burner to a direct-current data center is short, stiff and fast — provided the plasma is held steady and the load is followed on the electronics.

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Reproducibility

This paper is archived at doi:10.5281/zenodo.22645783 and its official version is maintained at <https://www.kronosfusionenergy.com/publications/>

[metrovolt-hvdc-native-power-interface.html](#). Every headline quantity is a frozen anchor (gates MV-L1-A1, MV-L1-A3, BU-L1-A10, BU-L1-A1) in the Kronos Physics De-Risking Register, doi:10.5281/zenodo.22133057. The figure-generation scripts recompute every data figure from the frozen anchors of table 1 and the equations of §3; the pipeline is deterministic under a fixed seed. No result depends on unpublished or proprietary data, and no economic quantity is computed anywhere.

Data availability

The frozen anchors underlying this work are stamped in the de-risking register [32] (doi:10.5281/zenodo.22133057) and regenerable from the public burner deposit doi:10.5281/zenodo.21746479; the direct-energy-conversion physics is deposited at doi:10.5281/zenodo.22645700. Economics, siting and service-level content are excluded from the public deposit by design.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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